

Problem 26.8

A small business pays you an annual profit at the end of each year for 20 years. The payments grow at an annual rate of 2.5%, the first payment is \$10,000. What is the present value of this stream of payments at the annual effective rate of interest 6%?

Solution

The payments form a geometrically increasing annuity-immediate. The first payment is

$$10000,$$

the annual growth rate is

$$g = 0.025,$$

and the annual effective interest rate is

$$i = 0.06.$$

For a geometrically increasing annuity-immediate, the present value is

$$PV = P \left(\frac{1 - \left(\frac{1+g}{1+i} \right)^n}{i - g} \right).$$

Substitute $P = 10000$, $g = 0.025$, $i = 0.06$, and $n = 20$:

$$PV = 10000 \left(\frac{1 - \left(\frac{1.025}{1.06} \right)^{20}}{0.06 - 0.025} \right).$$

Thus,

$$PV = 139734.7567.$$

Therefore, the present value is

$$\boxed{139734.76}.$$